

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 34

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Anchovy r-selected; blue whale k-selected (1) • r-selected: mature quickly, produce many young, disperse widely, short lives; adapted to respond rapidly to high death rates / unstable conditions (mice, locusts, greenfly, rabbits) (1) • k-selected: mature late, few young with parental care, long-lived; populations stable near carrying capacity (whales, elephants, rhinos) (1)	3
01.2	<i>1 mark for each point.</i> • Low reproductive rate means losses from hunting / habitat change cannot be replaced quickly (1) • An increase in death rate above the low birth rate causes a population crash (1) • Recovery takes decades (eg blue whale still far below pre-whaling numbers after the 1986 moratorium); r-selected species rebound within a few years once exploitation stops (1)	3
01.3	<i>1 mark.</i> • New population = starting population + births + immigrants - deaths - emigrants (1)	1
01.4	<i>1 mark for each point.</i> • The greatest rate of exploitation (harvest) possible without causing long-term population decline (1) • Equals the surplus production: highest when the population is well below carrying capacity where growth rate is fastest (middle of the sigmoid curve) (1) • At carrying capacity net growth is zero so any harvest reduces the population; harvesting above MSY causes decline (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 79-80; AQA spec 3.1.3.4.	

Question	Marking guidance	Marks
02.1	<i>1 mark for each point.</i> • In-situ: conserving species in their natural habitat with the whole community; ex-situ: away from the natural habitat, eg zoos, botanic gardens, seed banks, captive breeding (1) • Needed when conservation in the wild is impossible or insufficient to prevent extinction (habitat gone, population too small, poaching) (1)	2
02.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Is the wild population threatened? (1) • Is there a genetically diverse captive population? (1) • Is in-situ conservation failing? (1) • Is keeping a captive population realistic (habitat size, food, relationships, cost)? (1) • Is release likely to be successful now or in future (suitable habitat, threats removed)? (1)	4
02.3	<i>2 marks for each point (partial credit for a partly correct point).</i> • Blue whale: habitat size far too large to provide; food (huge quantities of krill / plankton) cannot be supplied; cost (2) • Large blue butterfly: complex inter-species relationship - larvae are taken into ant nests where they overwinter feeding on ant grubs; requires specific ant species and thyme food plant (2)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-34; AQA spec 3.1.2.3.3.	
03.1	<i>1 mark for each point.</i> • Surveys monitor photosynthesis rates, biomass, soil water content and the spread of pests / diseases so inputs are applied only where needed (1) • Yield mapping of different parts of a field controls fertiliser application in each area for future crops (1) • This maximises yield while avoiding wastage / reducing leaching and nutrient pollution (1)	3
03.2	<i>1 mark for each point.</i> • Machinery, pesticides, fertilisers, improved breeds and infrastructure (transport, refrigeration) are expensive (1) • They are mainly available in affluent societies; poorer farmers cannot afford the inputs (1)	2
03.3	<i>1 mark.</i> • Provides an income that allows producers to afford basic rights such as water, education, health care and food (1)	1
03.4	<i>1 mark for each point.</i> • Nitrates are leached into water bodies used as sources of potable water (1) • High nitrate levels cause blue baby syndrome (methaemoglobinaemia) and nitrates may be carcinogenic (1)	2
03.5	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Anaerobic digestion by microbes in rice padi fields (1) • Livestock intestines (ruminant digestion) (1) • Anaerobic decay of manure / slurry (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 329-332; AQA spec 3.5.1.3-3.5.1.4.	

Question	Marking guidance	Marks
04.1	<i>1 mark for each point.</i> • Quadrat size depends on the size / spacing of the organisms: 0.5 m x 0.5 m is typical for ground flora (1) • 10 cm x 10 cm is for lichens (too small to contain a representative sample of plants); 100 m x 100 m is for trees (impractical, too much work) (1)	2
04.2	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $N = 240$; $N(N-1) = 240 \times 239 = 57\,360$ (1) • $\Sigma n(n-1) = 48 \times 47 + 120 \times 119 + 36 \times 35 + 24 \times 23 + 12 \times 11 = 2256 + 14280 + 1260 + 552 + 132 = 18\,480$ (1) • $D = 57\,360 / 18\,480 = 3.10$ (accept 3.1) (1)	3
04.3	<i>1 mark for each point.</i> • Lower D means lower biodiversity: the ungrazed meadow is dominated by one or a few species (eg tall grass) with the others rare (less even) (1) • Grazing (plagioclimax management) prevents dominance by vigorous species so abundance is more evenly spread (1)	2
04.4	<i>1 mark for each point.</i> • A frame quadrat flattens tall plants, making percentage cover estimates inaccurate and hiding shorter plants (1) • Point quadrat pins are lowered through the vegetation; each touch counts (10 pins = 10% each) without flattening (1)	2
04.5	<i>1 mark.</i> • Subjective judgement varies between recorders; results are not quantitative so statistical analysis is not possible (but quick for many areas) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 394-395, 401-402; AQA spec 3.7.2.1.	
05.1	<i>1 mark for each point.</i> • South America: high loss ($5.1\text{--}5.5 \text{ Mha yr}^{-1}$) falling by about half by 2020 (1) • Africa: loss steady at about 3.4 then rising to 3.9 by 2020 - now the highest (1) • Europe: negative values - forest area increasing (afforestation exceeds loss), though the gain is shrinking (1)	3
05.2	<i>Up to 4 marks in total; award marks in proportion to the completeness of each point.</i> • South America: clearance for commercial cash crops (sugar cane, soya) and cattle ranching for export; HEP reservoirs; roads improving access; mining (up to 2) • Africa: growing rural populations clearing land for subsistence farming; fuelwood and charcoal collection above regrowth (charcoal wastes much of the wood's energy); fodder cutting in dry areas (up to 2) • Both: hardwood timber (teak, mahogany) from clear-felled mixed forest often not replanted; urban expansion (up to 2)	4
05.3	<i>1 mark for each point.</i> • They create barriers fragmenting wildlife populations (1) • Improved access makes it easier to remove timber and farm produce, so clearance spreads along them (1)	2

Question	Marking guidance	Marks
05.4	<i>1 mark.</i> Conifers grow quickly so plantations are replanted after harvest and yield a saleable crop within decades; slow-growing hardwoods (teak, mahogany) take too long for commercial plantations so they are taken from natural forest that is clear-felled and not replanted (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 367-368; AQA spec 3.5.3.3.	
06.1	<i>1 mark for each point.</i> - Giant tritons are predators of the crown-of-thorns starfish which eats coral polyps (1) - Removing tritons allowed starfish populations to explode, destroying large areas of reef (food web / trophic cascade) (1)	2
06.2	<i>1 mark for each point.</i> - Fixed mooring buoys so boats do not anchor on coral (1) - Education of divers / snorkellers not to touch coral or kick up sediment; bans on spear fishing and souvenir collection (1) - Fishing controls: no-take zones (Chagos, Great Barrier Reef, Lundy, Lamlash Bay), seasons, minimum and maximum catch sizes, bans on dynamite / small-mesh nets / trawling (1) - Protection of turtle nesting beaches; control of coastal development; protection of mangroves and seagrass; artificial reefs (reef balls, sunk ships / oil rigs) (1)	4
06.3	<i>1 mark for each point.</i> - No sunlight so no symbiotic algae; polyps rely only on captured food so have far less energy and grow very slowly (1) - Damage from trawls takes hundreds of years to repair; the slow-growing, low-fecundity fish (orange roughy) they support are easily over-exploited (1)	2
06.4	<i>1 mark for each point.</i> - Deep-water trawling (also by-catch); oil and gas exploration in deeper water; ocean acidification (1) - Protected areas banning trawling, eg Darwin Mounds SAC off NW Scotland, Oculina Bank off Florida (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 61-64; AQA spec 3.1.2.3.4.	
07.1	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Potato bean (N America): high protein content (1) - Moroma bean (southern Africa): protein content similar to soya, drought resistant (1) - Yeheb tree (Somalia): edible nuts, drought resistant, grows on poor soils (1) - Perennial maize (Mexico) / Kernza: no need to buy seed each year, less ploughing so less soil erosion (1) - American bison: graze grassland too poor for cattle (1) - Common eland: browse and graze so can be kept with cattle without severe competition (1)	2

Question	Marking guidance	Marks
07.2	<i>1 mark for each point.</i> - Indigenous species are better adapted to local conditions such as climate, pests / diseases and soil (1) - So they have higher survival rates / productivity is not reduced, as it is when introduced species are kept in areas to which they are not adapted (1)	2
07.3	<i>1 mark.</i> They grow in soils with a high salt content, so their genes could be used in breeding programmes to increase the salt tolerance of crops grown on soils salinised by irrigation (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 11; AQA spec 3.1.2.1.1.	
08.1	<i>1 mark for each point.</i> - Chicken and salmon need far less feed per kg gained (1.2-1.8 vs 7.0) (1) - Fish are ectotherms so no energy is used for thermoregulation and water supports their mass; poultry grow fast and are kept warm indoors with movement restricted (1) - Cattle are large endotherms, use energy in movement, and lose energy in methane from cellulose digestion (1)	3
08.2	<i>1 mark for each point.</i> - Control of movement (housing / restricted grazing) so less energy is used in muscle activity (1) - Temperature control (heated / insulated buildings) so less energy is used in maintaining body temperature (1) - Species selection: choose species with low FCRs (poultry, fish); high-energy easily digested diets; slaughter at the fastest growth stage (1)	3
08.3	<i>1 mark for each point.</i> - Only about 10% of the energy at one trophic level is passed to the next; most is lost as heat in respiration / metabolism or in undigested material (1) - Humans eating plants (primary consumers) receive more of the energy captured by photosynthesis than eating animals fed on those plants (secondary consumers) (1) - Livestock better where land cannot grow crops (upland, semi-arid, steep, thin soils): grazing animals with symbiotic gut bacteria digest cellulose in grass that humans cannot (1) - Omnivores (pigs) convert food wastes (dairy by-products, catering waste) into meat (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.	

Question	Marking guidance	Marks
09.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Size: shows whether the population is viable / how urgent action is; sets a baseline to judge success; needed to estimate carrying capacity and sustainable off-take (2) • Distribution: shows which habitats / areas are used and whether the population is fragmented into non-viable isolated groups needing corridors; identifies key areas to protect (2) • Survival rate (mortality): identifies the life stage and causes of death (predation, disease, poaching) so management targets the limiting factor (2) • Age structure: many young indicates a growing population; few young / ageing population indicates breeding failure or high juvenile mortality even if adults are numerous; k-selected species need breeding adults protected (2)	8
09.2	<i>1 mark for each point.</i> • Direct counts from vehicles / aircraft / drones along transects, or camera traps with individual recognition (1) • Sub-sample counts in known areas scaled up to the whole reserve; mark-release-recapture (Lincoln index) for smaller species (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-402; AQA spec 3.1.2.3.5.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Water: catchment management, conservation, metering, greywater, re-use of treated effluent, aquifer recharge, sustainable urban drainage - largely cyclical already; limits in arid regions • Minerals: exploiting low-grade ores with low-temperature processes (bioleaching), extending product lifespans, repair / re-use / repurpose / recycle, design for disassembly, reduced packaging; limits: dispersal, alloys, energy of collection • Waste management: repair, re-use, refurbish, recycle, composting and anaerobic digestion returning nutrients, minimal landfill, energy from residual waste; EU targets on packaging, batteries, WEEE • Pollution control: shift from post-production treatment to non-release - fuel desulfurisation not FGD, fuel cells / electric vehicles not combustion, industrial symbiosis (Kalundborg: waste heat, gypsum, fly ash, yeast used by neighbours); limits for legacy pollution • Energy: renewables, storage, energy density technologies, low-temperature manufacturing, CHP; limits of intermittency, transport fuels • Food: natural processes for nutrients (legumes, organic matter), pest control (IPM), soil biota, agri-environment schemes, IMTA aquaculture; tension with maximum yield for a growing population • Land uses supporting ecosystems: urban green infrastructure, gardens, permeable surfaces, habitat in redeveloped docks / lagoons • Judgement: high applicability in some sectors (water, waste), partial in others (minerals, energy for transport); needs policy support	9
	Model answer - Level 3 (9 marks)	
	Taking water, waste and pollution control: water supply is already close to circular, because the hydrological cycle returns what is used. Catchment management, conservation, metering, grey-water re-use, re-use of treated effluent, aquifer recharge and sustainable urban drainage close the loop further, with the main limit in arid regions where abstraction exceeds recharge. Waste management can apply the principles almost completely: repair, re-use and refurbishment extend product life, recycling returns materials, composting and anaerobic digestion return nutrients and yield biogas, residual waste supplies energy and landfill is minimised, driven by EU targets on packaging, batteries and WEEE; the limits are mixed materials that cannot be separated and the energy cost of collection.	

Question	Marking guidance	Marks
	Pollution control changes character under circular principles: instead of treating wastes after production - flue-gas desulfurisation, dispersants - the aim becomes non-release, through fuel desulfurisation, fuel cells and electric vehicles, and industrial symbiosis, as at Kalundborg where one plant's waste heat, gypsum, fly ash and yeast become neighbours' raw materials. Land use can support ecosystems with green infrastructure, gardens, permeable surfaces and habitat in redeveloped docks.	
	Applicability is high for water and waste because natural cycles already exist to copy; it is partial for pollution control because many processes still depend on combustion, and it needs policy support - targets, producer responsibility, planning - to overcome the economics of linear systems. The principles can be applied widely, but not evenly.	
	Weaker answer - Level 1 (3 marks)	
	The circular economy means reusing and recycling things instead of throwing them away. For water this means saving water and reusing grey water. For waste management it means recycling more and putting less in landfill. For pollution control it means not producing pollution in the first place. The circular economy can be applied to all of these to make them more sustainable.	
	What makes the difference	
	• Level 1 gives one idea per area; Level 3 gives several specific measures for each and names the limit in each case.	
	• Level 3 uses a case study (Kalundborg) and policy examples (EU targets) to show application, not just principle.	
	• 'Evaluate the extent' - Level 3 compares applicability across the three areas and explains the differences; Level 1 says it applies to all.	
10.2	<i>1 mark.</i> • Locating industries so that the wastes of one (heat, gases, by-products) become the raw materials of another, mimicking natural cycles (eg Kalundborg eco-industrial park) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 375-379; AQA spec 3.6.4.1.	

Question	Marking guidance	Marks
11.1	Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Physical features: mass / gravity retaining an atmosphere and giving pressure for liquid water; distance from the Sun and the Goldilocks zone; speed of rotation reducing temperature extremes; tilt producing seasons; magnetic field deflecting the solar wind; the Moon stabilising the orbit and producing tides • Importance of liquid water: solvent, transport, temperature control, anomalous expansion, high specific heat capacity, habitats, absorption of UV before the ozone layer • Development of life: Archaea at vents; photosynthesis; oxygen reacting with iron then accumulating; ozone layer allowing colonisation of land; aerobic organisms • Life changing conditions: carbon sequestration offsetting solar brightening, biogeochemical cycles, transpiration and rainfall, regulation of atmospheric gases • Methods of studying past conditions: proxy data, isotope ratios, ice cores, Mg:Ca ratios; understanding of continental drift and ocean currents • Links to current threats (climate change, ozone depletion) and a justified conclusion about the interdependence of physical and biological factors Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-9; AQA spec 3.1.1.	25

Question	Marking guidance	Marks
11.2	Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Ecological features: cnidarian polyps, cilia, nematocysts, zooxanthellae symbiosis, limestone skeletons building reefs, complex structure and food web; specific abiotic requirements (temperature, light, salinity, low turbidity) • Importance: quarter of marine species, fisheries, medicines, carbon sink, erosion protection, tourism • Vulnerability: narrow abiotic tolerance means sensitivity to warming, acidification, sedimentation, nutrient pollution, oil, bleaching; slow growth so slow recovery; physical damage; overfishing and food-web effects (crown of thorns); introduced species; dependence on mangroves and seagrass • Conservation: MPAs, No Take Zones, fishing controls, tourism management, protection of linked habitats, CITES • Comparison with other habitats (rainforests, mangroves, Antarctica) where relevant • Evaluation of local vs global threats and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-62; AQA spec 3.1.2.3.4.	25